ELECTRIC VEHICLE (EV) DATA ANALYSIS

A Tableau Dashboard Project Report

Market Trends • Adoption Patterns • Manufacturer Performance

Presented by Asrarul Haque

AGENDA

What This Report Covers

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Project Overview & Objective

Why this dashboard was built

02

Business Problem & Dataset

Questions answered and data used

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Insights & Recommendations

What the data reveals and suggests

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PROJECT OVERVIEW

Objective of the Dashboard

Project Title

Electric Vehicle (EV) Data Analysis Dashboard using Tableau

The objective of this project is to analyze Electric Vehicle (EV) adoption across different states, manufacturers, model years, and vehicle types. The dashboard helps stakeholders understand EV market trends, identify top-performing brands, and support data-driven policy and business decisions.

1,50,413

Total Vehicles

77.6%

BEV Share

67.83 mi

Avg. Range

BUSINESS PROBLEM

Questions the Dashboard Answers

The electric vehicle market is growing rapidly. Government agencies, manufacturers, and analysts need fast, reliable answers — without manually analyzing large datasets.

?

How many EVs are registered?

?

Which manufacturers dominate the market?

?

Which states have the highest EV adoption?

?

How has EV adoption changed over the years?

?

What is the BEV vs. PHEV split?

?

Which models are the most popular?

DATASET

Fields Used for the Analysis

The dashboard uses Electric Vehicle registration data containing the following fields:

- **Vehicle ID**
- **Model**
- **Make**
- **Model Year**
- **EV Type**
- **Electric Range**
- **State**
- **CAFV Eligibility**
- **Vehicle Category**

DATA PREPARATION

Getting the Data Analysis-Ready

1. Imported the dataset into Tableau
2. Removed duplicate records
3. Handled missing values
4. Standardized manufacturer and state names
5. Verified electric range values
6. Checked data types
7. Created calculated fields where required

TOOLS & TECHNIQUES

Tableau Features Used

- Worksheets
- Dashboards
- Filters
- Calculated Fields
- Parameters
- KPI Cards
- Line Chart
- Filled Map
- Bar Chart
- Donut Chart
- Top N Analysis
- Dashboard Actions

HEADLINE METRICS

KPI Cards

1,50,413

Total Vehicles

Total number of electric vehicles in the dataset

67.83 mi

Avg. Electric Range

Average electric driving range across all EVs

1,16,745

Total BEV (77.6%)

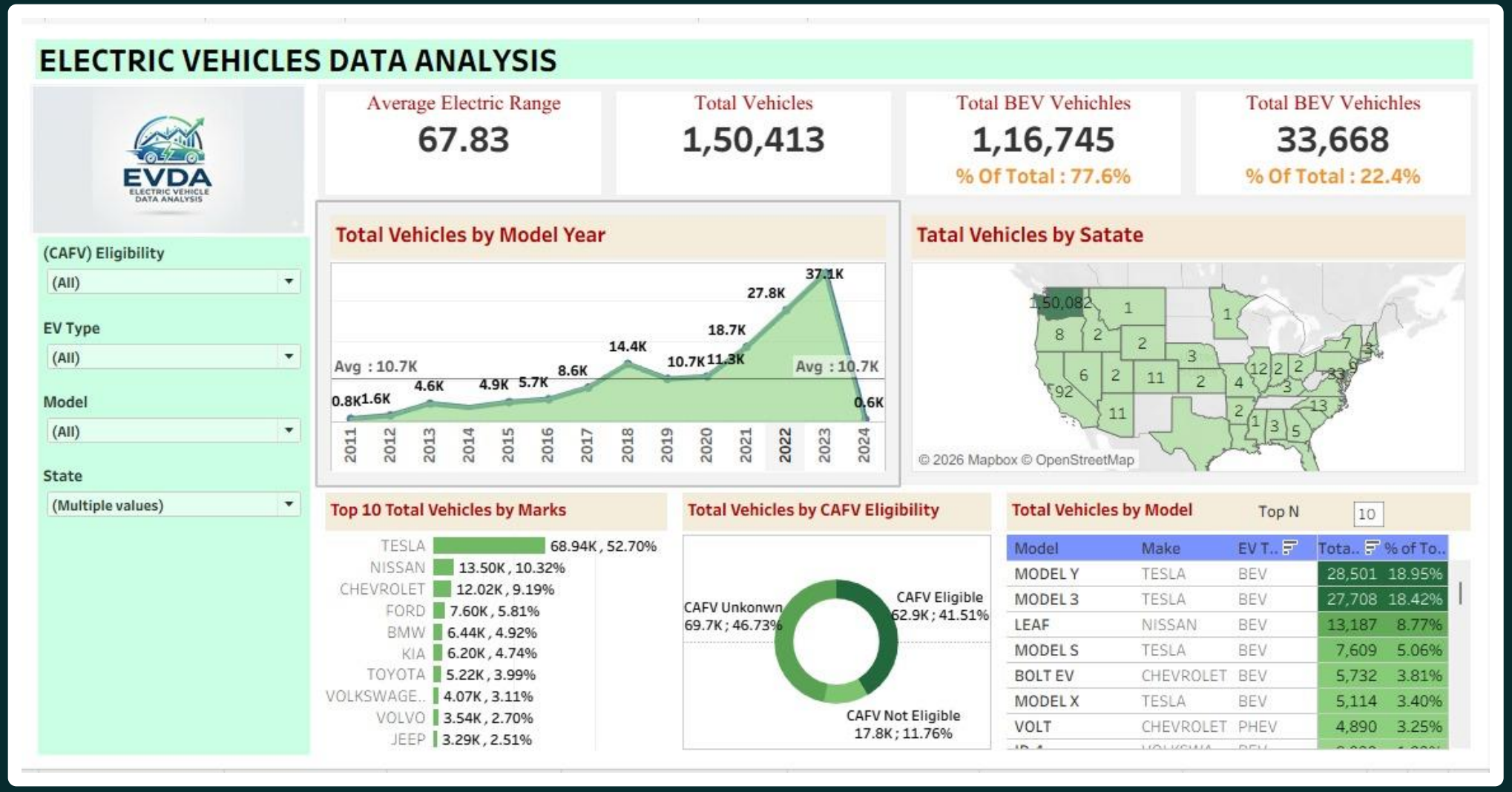
Battery Electric Vehicles — count & share

33,668

Total PHEV (22.4%)

Plug-in Hybrid Electric Vehicles — count & share

EV Data Analysis — Full View



DASHBOARD VISUALIZATIONS

Trends & Geography

↗ Total Vehicles by Model Year

Line chart displaying EV registrations by model year, from 2011 through 2024.

BUSINESS VALUE

Helps identify EV adoption trends and market growth over time — registrations rose sharply from ~10.7K average toward a 2023 peak of 37.1K.

◉ State-wise Vehicle Distribution

A filled map showing the number of EVs registered in each U.S. state.

BUSINESS VALUE

Identifies regions with high and low EV adoption to guide infrastructure and policy focus, with one leading state showing 1,50,082 registrations.

DASHBOARD VISUALIZATIONS

Manufacturers, Eligibility & Models

Top 10 Manufacturers

Ranks manufacturers — Tesla, Nissan, Chevrolet, Ford, BMW, Kia, Toyota.

BUSINESS VALUE

Shows market leaders and supports competitive analysis. Tesla leads with 68.94K vehicles (52.7%).

CAFV Eligibility

Splits vehicles into CAFV Eligible, Not Eligible, and Unknown categories.

BUSINESS VALUE

Helps evaluate compliance with clean vehicle incentive programs across the fleet.

Top Vehicle Models

Lists the highest-selling EV models, led by Tesla Model Y and Model 3.

BUSINESS VALUE

Identifies the most popular electric vehicle models to inform strategy.

USER INTERACTIVITY

Interactive Filters

FILTER

CAFV Eligibility

Analyze vehicles based on eligibility status

FILTER

EV Type

Compare BEV and PHEV vehicles side by side

FILTER

Model

Filter the dashboard by a specific vehicle model

FILTER

State

Analyze adoption patterns in selected states

FINDINGS

Key Insights

- Total EV registrations exceed 1,50,000 vehicles.
- BEVs represent about 77.6% of all vehicles; PHEVs account for 22.4%.
- The average electric range across the fleet is 67.83 miles.
- EV adoption has grown significantly in recent model years, peaking in 2023.
- Tesla is the leading manufacturer by vehicle count (52.7% share).
- Tesla Model Y and Model 3 are among the top-performing models.
- EV adoption varies considerably across states, showing uneven market penetration.

STRATEGY

Business Recommendations

- → Expand EV infrastructure in states with lower adoption.
- → Focus marketing efforts on regions showing rapid growth.
- → Encourage incentives for PHEVs and other emerging EV categories.
- → Analyze the factors driving Tesla's market leadership to inform competitive strategy.
- → Continue monitoring model-year trends to forecast future demand.

REFLECTION & ROADMAP

Challenges Faced & Future Improvements

CHALLENGES FACED

- Cleaning a large EV dataset
- Designing a dashboard with multiple interactive filters
- Creating a clear geographic visualization
- Presenting multiple KPIs without clutter
- Ensuring dashboard responsiveness

FUTURE IMPROVEMENTS

- Sales forecasting
- Battery capacity analysis
- State-wise growth forecast
- Interactive tooltips & drill-downs
- Charging station analysis
- Manufacturer growth trends
- Customer demographics
- Dynamic parameters



Thank You

Electric Vehicle (EV) Data Analysis Dashboard — Tableau Project

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